

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills		
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Date:	03/08/2026	Duration:	60 minutes

Code of Conduct

I S.GAYATHERI(192521306) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: _____

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. **Study and Analyze the Scenario** ○ Carefully understand the given problem statement.
 - Identify the business domain, stakeholders, existing challenges, and system objectives.
2. **Identify Functional and Non-Functional Requirements** ○ List all functional requirements required by the proposed system.
 - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.

3. **Identify Actors and Use Cases**
 - Identify all primary and secondary actors interacting with the system.
 - List the major use cases associated with each actor.
 - Draw a Use Case Diagram representing the interactions between actors and the system.
4. **Prepare the Software Requirement Specification (SRS)**
 - Develop an SRS document containing:
 - Introduction
 - Problem Description
 - Scope of the System
 - Functional Requirements
 - Non-Functional Requirements
 - Use Case Descriptions
 - Data Requirements
 - External Interface Requirements
 - System Features
5. **Identify Assumptions and Constraints**
 - List the assumptions considered while developing the system.
 - Identify technical, operational, economic, legal, and time-related constraints affecting the project.
6. **Validate Requirement Completeness and Consistency**
 - Verify whether all stakeholder requirements have been addressed.
 - Check for ambiguity, redundancy, inconsistency, and missing requirements.
 - Suggest improvements to enhance the quality and completeness of the requirement specification.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modeling	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.

Software Requirement Specification	SRS is well-structured, complete, professionally	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or
(SRS) Documentation	organized, and includes all required sections.			lacks essential components.
Assumptions, Constraints, and Requirement Validation	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	Mark
Scenario Understanding and Problem Analysis	/5
Functional and Non-Functional Requirement Identification	/5
Actor Identification and Use Case Modeling	/5
Software Requirement Specification (SRS) Documentation	/5
Assumptions, Constraints, and Requirement Validation	/5
TOTAL	/5

CO2 - ASSESSMENT TOOL – 01

Q29. AI-Powered Personalized Nutrition Recommendation System

1.1 Business Domain

The proposed system belongs to the Healthcare, Nutrition, and Artificial Intelligence (AI) domain. It is designed to provide personalized nutrition recommendations by analyzing an individual's health information, lifestyle, dietary preferences, and medical history using Artificial Intelligence. In recent years, healthcare organizations have increasingly adopted AI technologies to improve patient care, reduce manual workload, and provide data-driven healthcare solutions. The AI-Powered Personalized Nutrition Recommendation System aims to replace traditional one-size-fits-all diet plans with customized nutritional guidance tailored to each individual's specific health requirements. By considering various factors such as age, gender, body mass index (BMI), physical activity level, allergies, chronic diseases, and nutritional goals, the system generates scientifically balanced meal plans that promote healthy living.

1.2 Stakeholders

Stakeholders are the individuals or organizations that directly or indirectly interact with or benefit from the proposed system. The primary stakeholder is the user or patient, who provides personal and health-related information to receive customized nutrition recommendations. Users can also monitor their daily food intake, track nutritional progress, and receive health insights generated by the AI system.

Another important stakeholder is the dietitian or nutritionist, who uses the system to review user health information, verify AI-generated meal plans, modify nutritional recommendations, and provide expert dietary advice whenever necessary. The platform assists dietitians by reducing the time required to manually analyze nutritional requirements while improving the accuracy of recommendations.

1.3 Existing Challenges

Despite the availability of many diet-planning applications, several challenges continue to affect the effectiveness of nutritional management. One of the major problems is that generic diet plans do not satisfy individual nutritional requirements. Most existing systems provide the same dietary recommendations to all users without considering differences in age, body weight, health conditions, allergies, or activity levels. As a result, the recommended diets may not be suitable for every individual and may fail to achieve the desired health outcomes.

Another significant challenge is the lack of personalized nutrition recommendations. Every individual has unique nutritional needs based on their medical history, metabolism, lifestyle, and health goals. Existing systems often fail to analyze these factors comprehensively, leading to inaccurate or ineffective dietary guidance.

1.4 System Objectives

The primary objective of the proposed AI-Powered Personalized Nutrition Recommendation System is to develop an intelligent healthcare platform that provides customized nutrition plans for every individual based on their personal health information. The system aims to eliminate the limitations of generalized diet plans by generating personalized recommendations that match each user's unique nutritional requirements.

The system analyzes various health parameters, including age, gender, height, weight, Body Mass Index (BMI), physical activity level, medical history, allergies, dietary preferences, and existing health conditions. Using Artificial Intelligence algorithms, the platform processes this information to determine the user's calorie requirements, nutrient distribution, and suitable food choices.

2. Functional Requirements

Functional requirements describe the specific services and operations that the AI-Powered Personalized Nutrition Recommendation System must perform to meet the needs of its users and stakeholders. These requirements define how the system should collect health information, process user data, generate personalized nutrition recommendations, monitor dietary intake, and assist healthcare professionals in providing better nutritional guidance.

2.1 User Management

The system shall provide a secure user management module that enables individuals to create and manage their accounts. New users should be able to register by providing personal information such as their name, email address, phone number, date of birth, and password. After successful registration, users can log into the system using authenticated credentials. Passwords should be securely encrypted to protect user privacy. The system should also provide a password recovery option through email or mobile verification, allowing users to regain access if they forget their login credentials.

2.2 Health Data Management

The health data management module is responsible for collecting and maintaining all information required for generating personalized nutrition recommendations. Users should be able to enter details such as age, gender, height, weight, Body Mass Index (BMI), blood group, medical history, allergies, chronic diseases, medications, dietary preferences, and physical activity levels. The system should validate the entered information before storing it in a secure database to ensure data

accuracy and reliability. Users should also have the option to update their health records whenever there is a change in their medical condition or lifestyle.

2.3 AI-Based Nutrition Recommendation

One of the most important functionalities of the proposed system is the Artificial Intelligence recommendation engine. The AI engine should analyze the user's health profile, medical history, nutritional requirements, activity level, and dietary preferences to generate customized meal plans. The system should calculate daily calorie requirements and recommend balanced meals containing appropriate amounts of carbohydrates, proteins, fats, vitamins, minerals, fiber, and water intake. It should also avoid recommending foods that may cause allergic reactions or negatively affect existing medical conditions such as diabetes, hypertension, kidney disease, or obesity.

2.4 Nutrition Tracking

The nutrition tracking module allows users to monitor their daily food consumption and evaluate whether they are meeting their nutritional goals. Users can record their breakfast, lunch, dinner, snacks, and beverages consumed throughout the day. The system automatically calculates total calorie intake along with the amounts of proteins, carbohydrates, fats, vitamins, minerals, and water consumed. It compares the actual nutritional intake with the recommended values generated by the AI engine and highlights any nutritional deficiencies or excess consumption.

2.5 Health Monitoring

The health monitoring module continuously evaluates the user's nutritional progress and overall health status. The system compares actual nutritional intake with personalized dietary goals and monitors changes in weight, BMI, calorie consumption, and other health indicators over time. Weekly and monthly reports are automatically generated to summarize nutritional performance and identify trends in eating habits. Based on these reports, the system provides personalized health insights and recommendations to encourage healthier lifestyle choices..

2.6 Notification Management

The notification module helps users maintain consistency in following their nutrition plans by sending automated reminders and alerts. The system should notify users about meal timings, water intake, medication schedules (if applicable), and nutritional goals throughout the day. Progress notifications should inform users about achievements such as meeting calorie targets or improving nutritional scores. Health alerts should also be generated whenever unusual dietary patterns or nutritional deficiencies are detected. These reminders encourage users to develop healthy habits and improve long-term adherence to their personalized nutrition plans.

2.7 Administrator Management

The administrator module provides complete control over the operation and maintenance of the system. Administrators should be able to manage user accounts, verify healthcare professionals, maintain the food and nutrition database, update nutritional information, monitor AI recommendation performance, and generate administrative reports. They are also responsible for monitoring system security, maintaining server performance, performing database backups, resolving technical issues, and ensuring continuous availability of the application.

3. Non-Functional Requirements

Non-functional requirements define the quality standards that the AI-Powered Personalized Nutrition Recommendation System must satisfy to ensure efficient, secure, reliable, and user-friendly operation. These requirements focus on the performance, security, usability, scalability, maintainability, and availability of the application rather than its specific functionalities.

3.1 Performance

The system should provide fast and efficient performance by generating personalized nutrition recommendations within three seconds after processing user health information. It should support at least 5,000 concurrent users without affecting response time or system stability. Database operations, report generation, and AI analysis should be optimized to ensure smooth user experience even during peak usage. Efficient resource utilization and cloud-based infrastructure should be employed to maintain high system performance.

3.2 Security

Since the platform stores highly sensitive personal and medical information, security is one of the most critical requirements. The system should implement secure user authentication, encrypted password storage, and encrypted communication using HTTPS protocols. Role-based access control should be implemented to ensure that only authorized users can access specific information. Healthcare professionals should access only their assigned patient records, while administrators manage system-level operations. Regular security audits, data backup, and protection against cyber threats should also be incorporated to safeguard confidential health information.

3.3 Reliability

The proposed system should provide reliable service with 99.9% availability, ensuring that users can access personalized nutrition recommendations whenever required. Automatic database backup mechanisms should prevent data loss during unexpected failures, while recovery procedures should restore system functionality with minimal downtime. The application should continue operating smoothly under different workloads and recover gracefully from hardware or software failures.

3.4 Usability

The system should provide an intuitive, attractive, and user-friendly interface that can be easily understood by users with varying levels of technical knowledge. Navigation should be simple, and important features such as health profile management, meal tracking, and nutrition reports should be easily accessible. The application should support both mobile devices and web browsers, allowing users to conveniently access the platform from smartphones, tablets, and computers. Clear instructions, icons, and graphical reports should further improve the user experience.

3.5 Scalability

The AI-powered nutrition platform should be designed to accommodate future growth without requiring major architectural changes. As the number of users increases, the system should continue to deliver consistent performance. The food database should support the addition of new food items, nutritional values, and dietary categories. The modular architecture should also allow future integration with wearable fitness devices, hospital information systems, electronic health records, and advanced AI models.

3.6 Maintainability

The software should be developed using a modular architecture that allows developers to easily modify, update, and maintain individual system components without affecting the entire application. Software updates, bug fixes, AI model improvements, and database modifications should be implemented efficiently with minimal interruption to users. Proper documentation, coding standards, and testing procedures should be followed to simplify future maintenance activities and ensure long-term software quality.

3.7 Availability

The AI-Powered Personalized Nutrition Recommendation System should be available 24 hours a day and 7 days a week so that users and healthcare professionals can access personalized nutrition services whenever required. Cloud-based deployment should ensure uninterrupted service, automatic failover mechanisms, and disaster recovery support. Continuous monitoring of servers and network infrastructure should help maintain system availability and provide reliable access to health information from any location with an internet connection.

3.1 Actors

Actors are the individuals or external systems that interact with the AI-Powered Personalized Nutrition Recommendation System to perform specific tasks. They represent the different users and supporting services that contribute to the overall functionality of the system. The proposed system

includes both primary actors, who directly interact with the application, and secondary actors, which provide supporting services to ensure the smooth operation of the platform.

Primary Actors

The User (Patient) is the main actor of the system. The user registers an account, logs into the application, enters personal and health-related information, receives AI-generated personalized meal plans, records daily food intake, monitors nutritional progress, and views health reports. Users also receive notifications and reminders that help them maintain healthy eating habits and achieve their nutritional goals.

The Dietitian or Nutritionist is responsible for reviewing the user's health profile and AI-generated meal plans. Dietitians can modify nutrition recommendations based on medical conditions, allergies, and specific dietary needs.

Secondary Actors

The AI Recommendation Engine acts as the intelligent decision-making component of the system. It analyzes user health information, calculates nutritional requirements, predicts calorie needs, and generates personalized meal plans based on nutritional guidelines and user preferences. The AI engine continuously improves its recommendations using updated health information and user feedback.

The Notification Service supports the application by sending automated reminders and alerts to users. These notifications include meal reminders, water intake reminders, health alerts, medication reminders (if applicable), and progress updates. This service encourages users to follow their nutrition plans consistently.

3.2 Use Cases

A use case describes a specific interaction between an actor and the system to accomplish a particular objective. The AI-Powered Personalized Nutrition Recommendation System provides several use cases that enable users and healthcare professionals to effectively manage nutritional information.

The User performs activities such as registering a new account, logging into the system, updating personal information, entering health details, requesting personalized meal plans, recording daily meals, monitoring calorie and nutrient intake, viewing nutritional reports, and receiving automated notifications. These functions allow users to maintain healthy eating habits and continuously monitor their nutritional progress.

3.3 Use Case Diagram (Explanation)

The Use Case Diagram illustrates how different actors interact with the AI-Powered Personalized Nutrition Recommendation System. The User is connected to use cases such as Register, Login, Update Profile, Enter Health Data, View Personalized Meal Plan, Track Meals, View Reports, and Receive Notifications. These interactions represent the core functionalities available to every registered user.

The Dietitian interacts with the system by reviewing user health information, updating personalized meal plans, and approving nutrition recommendations. The Doctor accesses nutritional reports, monitors patient progress, and recommends dietary modifications that support medical treatment..

3.4 Text Representation of Use Case Diagram

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| AI-Powered Personalized Nutrition System |
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User ----- Register

----- Login

----- Manage Profile

----- Enter Health Information

----- View Personalized Meal Plan

----- Track Daily Nutrition

----- View Health Reports

----- Receive Notifications

Dietitian ----- Review User Health Data

----- Modify Meal Plan

----- Approve Nutrition Plan

Doctor ----- Review Patient Nutrition

----- Monitor Health Progress

----- Recommend Dietary Changes

Administrator ----- Manage Users
----- Manage Food Database
----- Generate Reports
----- System Maintenance

AI Recommendation Engine -- Analyze Health Data
-- Generate Personalized Meal Plan

Notification Service ----- Send Meal Reminders
-- Send Water Reminders
-- Send Health Alerts

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Part 4

Part 4: Software Requirement Specification (SRS)

4.1 Introduction

The AI-Powered Personalized Nutrition Recommendation System is an intelligent healthcare application developed to provide customized nutritional guidance based on an individual's health profile, medical history, dietary preferences, and physical activity level. Unlike traditional nutrition systems that provide generalized diet plans, this system utilizes Artificial Intelligence (AI) to analyze user-specific health information and generate personalized meal recommendations.

4.2 Problem Description

Many existing nutrition applications provide standard diet plans that fail to consider the unique nutritional requirements of each individual. Factors such as age, gender, body weight, body mass index (BMI), medical history, allergies, chronic diseases, lifestyle, and physical activity significantly influence nutritional needs. Generic meal plans often produce poor health outcomes because they do not address these individual differences.

Healthcare professionals spend considerable time manually evaluating patient information before preparing personalized nutrition plans. This manual process is time-consuming, increases workload, and may result in inconsistencies.

4.3 Scope of the System

The scope of the proposed system includes the complete management of personalized nutrition planning and monitoring. The system collects user health information such as age, gender, height, weight, BMI, medical history, allergies, dietary preferences, and physical activity levels. Based on this information, the Artificial Intelligence engine analyzes nutritional requirements and generates personalized meal plans suitable for the user's health condition.

The system also allows users to record their daily meals and automatically calculates calorie intake and nutrient consumption. It compares the actual nutritional intake with recommended dietary goals and generates progress reports that help users improve their eating habits. Healthcare professionals can review these reports and modify recommendations whenever necessary.

4.4 Functional Requirements

The AI-Powered Personalized Nutrition Recommendation System includes several functional modules that work together to deliver personalized nutritional guidance. The system enables users to register securely, log in using authenticated credentials, and maintain personal health profiles. Users can enter health information such as medical history, allergies, body measurements, physical activity levels, and dietary preferences.

The Artificial Intelligence engine processes the collected information and generates personalized meal recommendations based on nutritional guidelines and user-specific health conditions. The system recommends calorie intake, nutrient distribution, suitable food items, and water consumption while avoiding foods that may trigger allergies or worsen existing medical conditions.

4.5 Non-Functional Requirements

The system must provide high performance by generating personalized meal recommendations within three seconds of receiving user input. It should support thousands of concurrent users without affecting response time or application stability. Strong security measures such as encrypted passwords, secure communication protocols, role-based access control, and protected health data storage must be implemented to safeguard sensitive information.

The platform should provide reliable operation with 99.9% availability and automatic backup mechanisms to prevent data loss. A user-friendly interface should enable individuals of different age groups and technical backgrounds to easily navigate the application.

4.6 Use Case Descriptions

Use Case 1: User Registration

Actor: User

Description: The user registration process allows new users to create an account in the system by providing personal information such as name, email address, mobile number, and password. The

system validates the entered information, verifies that the email or phone number is unique, encrypts the password, and stores the account details securely in the database. After successful registration, the user can log in and access personalized nutrition services.

Precondition: The user must not already have an account.

Postcondition: A new user account is successfully created and stored in the database.

Use Case 2: Generate Personalized Meal Plan

Actor: User

Description: After logging into the system, the user enters health information including age, height, weight, BMI, allergies, dietary preferences, medical history, and physical activity level. The Artificial Intelligence engine analyzes this information and calculates the user's nutritional requirements. Based on the analysis, the system generates a personalized meal plan that includes calorie recommendations, nutrient distribution, suitable food items, meal timings, and water intake. The user can view and follow the recommended nutrition plan.

Precondition: User profile and health information must be available.

Postcondition: Personalized meal recommendations are generated and displayed to the user.

Use Case 3: Nutrition Tracking

Actor: User

Description: The nutrition tracking module enables users to record their daily meals throughout the day. The system automatically calculates calorie intake, proteins, carbohydrates, fats, vitamins, minerals, and water consumption. It compares the actual nutritional intake with the recommended values and generates graphical reports showing nutritional progress. The system also identifies nutritional deficiencies and provides recommendations to improve dietary habits.

Precondition: Personalized meal plan has already been generated.

Postcondition: Daily nutritional records are stored and progress reports are updated.

4.7 Data Requirements

The proposed system requires accurate and well-organized data to generate reliable nutrition recommendations. User information such as name, age, gender, height, weight, BMI, medical history, allergies, dietary preferences, medications, and physical activity levels must be collected and securely stored. The system also maintains a comprehensive food database containing food names, calorie values, proteins, carbohydrates, fats, vitamins, minerals, fiber content, and serving sizes..

4.8 External Interface Requirements

The application provides a simple and user-friendly interface that can be accessed through both mobile applications and web browsers. The system supports smartphones, tablets, laptops, and desktop computers connected to the internet. It integrates with the AI Recommendation Engine for nutritional analysis, a centralized database server for storing health records, and a notification service for sending meal reminders and health alerts. Communication between the user and the system is secured using HTTPS protocols, while email and SMS services are used for account verification and notifications.

4.9 System Features

The AI-Powered Personalized Nutrition Recommendation System provides several advanced features that improve nutrition management and healthcare services. The system generates AI-based personalized meal recommendations according to individual health conditions and dietary preferences. It includes secure user profile management, daily nutrition tracking, calorie and nutrient monitoring, progress analysis, and detailed health reports. The platform also supports healthcare professionals by allowing dietitians and doctors to review patient nutrition records and modify meal plans whenever necessary. Automated reminders encourage users to maintain healthy eating habits and adequate hydration.